

Synthesis Characterization and Physicochemical Properties of Rigid Alicyclic Polyimide Films Based on Bicyclo[2.2.2]oct-7-ene-2,3,5,6-tetracarboxylic Dianhydride

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Abstract

Four polyimides based on bicyclo[2.2.2]oct-7-ene-2,3,5,6-tetracarboxylic dianhydride (BTD) with different rigid substituted diamines were synthesized: BTD-MIMA, BTD-HFA, BTD-FND, and BTD-TPM. These polyimides were soluble in organic solvents and had molecular weights between 39 and 70 kDa. Thermal stability above 400 °C and high glass transition temperatures between 272 and 355 °C were observed. Membrane films showed gas permeability properties suitable for gas separation applications.

2. Experimental

Polyimides were synthesized via two-step polycondensation in NMP at room temperature. BTD dianhydride (5 mmol) was dissolved in NMP; diamine (5 mmol) was added and stirred 24 h at 25 °C to form poly(amic acid). Chemical imidization was performed using acetic anhydride and triethylamine. Molecular weights were determined by GPC in DMF/LiBr. DSC measurements were made at 10 °C/min under nitrogen (second heating scan). TGA was performed at 10 °C/min under nitrogen and air.

3. Results and Discussion

The alicyclic BTD unit significantly improved solubility compared to fully aromatic PIs. The rigid bicyclic skeleton restricts backbone flexibility, resulting in high Tg values despite the alicyclic character. BTD-MIMA showed the highest Tg (355 °C) due to the rigid methylenedianiline-based diamine MIMA contributing to chain stiffness.

Table 3. Molecular weights and thermal properties of BTD-based polyimides.

Sample	Diamine	Mn (kDa)	Mw (kDa)	PDI	Tg (°C) (DSC)	Td5% N ₂ (°C)	Td5% air (°C)	Char 800°C (%)
BTD-MIMA	MIMA	68	148	2.18	355	430	408	52
BTD-HFA	HFA	55	121	2.20	320	418	395	44
BTD-FND	FND	43	98	2.28	295	412	388	48
BTD-TPM	TPM	39	86	2.21	272	405	382	41

Note: Mn = number-average molecular weight; Mw = weight-average molecular weight; PDI = polydispersity index; Tg = glass transition temperature; Td5% = temperature of 5% weight loss. BTD = bicyclo[2.2.2]oct-7-ene-2,3,5,6-tetracarboxylic dianhydride; MIMA = 4,4'-methylenedianiline; HFA = 4,4'-(hexafluoroisopropylidene)dianiline; FND = 9,9-bis(4-aminophenyl)fluorene; TPM = 1,1,1-tris(4-aminophenyl)ethane.

3.3 Gas Transport Properties

Membrane films from BTD-MIMA and BTD-HFA exhibited the highest gas permeability. CO₂ permeability: BTD-MIMA = 18.2 Barrer, BTD-HFA = 14.6 Barrer at 35 °C, 3 bar. O₂/N₂ selectivity: BTD-MIMA = 4.1, BTD-HFA = 3.8. The free volume fraction correlated well with the Tg values (R² = 0.94).